# Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

## PROFESSIONAL SUMMARY

Staff AI engineer with **production healthcare AI and HIPAA compliance experience** who built two AI systems directly relevant to Rula's clinical AI mission. **AgentProbe** - open-source LangGraph platform with multi-agent orchestration (Orchestrator, RAG Tracer, Auto Fix), RAG pipelines (ChromaDB), multi-provider LLM routing (Anthropic, OpenAI, Groq), and a 5-metric evaluation framework with CI/CD regression gates. **AI on FHIR** - HIPAA-compliant clinical NLP system (BERT, spaCy, embeddings) for patient record queries, delivering ~60% retrieval time reduction. 8+ years Python production engineering; AWS (S3, EC2, RDS, Glue); PHI-compliant data pipelines at scale. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

## CORE COMPETENCIES

Multi-Agent Orchestration (LangGraph) | RAG Pipelines & Vector Search | LLM Evaluation Frameworks | LLM Integration (Anthropic, OpenAI, Groq) | Healthcare AI & HIPAA Compliance | Clinical NLP (BERT, spaCy, FHIR) | Python (8+ years production) | AWS (S3, EC2, RDS, Glue) | Observable AI Systems (CI/CD Eval Gates) | Production at Scale (1M+ records) | FastAPI / Docker / GitHub Actions | Staff-Level Architecture & Mentoring

## EXPERIENCE

### Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

**Software Engineer → Associate Software Engineer → Trainee**

Pune, India

- Achieved **100% PHI compliance** designing Libsodium encryption and AWS S3 secure pipelines for all resident health data - **zero security incidents** across 100K+ users. Direct analogue to HIPAA-regulated clinical AI at Rula.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data requiring output validation and quality monitoring.
- Built and delivered **20+ full-stack backend modules** (Python, PostgreSQL, JavaScript) across **120+ Agile sprints**; led Docker containerization for 30%+ of microservices; cut execution time by **50%**.
- Set technical direction through peer code reviews and secure coding standards; built **5+ internal debugging tools** adopted across squads - reducing resolution time by **40%**.
- Improved production query performance by **30%** across 200+ SQL queries; maintained **99.5% uptime** for 6+ consecutive months via TDD/BDD.

### George Mason University

Aug 2025 - May 2026

**Graduate Teaching Assistant - Python & Databases**

Fairfax, Virginia

- Mentored 30+ students in Python and databases; set quality standards and drove iterative improvement - the same mentoring and technical direction-setting Rula's Staff AI role requires.

## PROJECTS

### AgentProbe: Multi-Agent Orchestration Platform with Eval Framework OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Production platform delivering exactly what Rula's AI foundation requires: RAG, multi-agent orchestration, LLM integrations, and evaluation frameworks - built end-to-end as open-source. **Multi-agent orchestration**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with multi-step reasoning, tool calls, and inter-agent state management - directly maps to patient conversation, provider decision support, and clinical ops agents. **RAG pipeline**: ChromaDB + embeddings + retrieval quality scoring across multi-step agent runs. **LLM integrations**: dynamic routing across Anthropic, OpenAI, Groq, Ollama - selection by task type, latency, and capability. **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with regression tracking and CI/CD gates - observable, safe AI that catches quality drift before it reaches users. **Observability**: 15-endpoint FastAPI backend + React dashboard.
*github.com/venkatalakshmiparimala/AgentProbe*

### AI on FHIR: HIPAA-Compliant Clinical NLP for Patient Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP system for clinical decision support in a regulated healthcare environment - the exact domain Rula operates in. **Clinical NLP**: BERT + spaCy + embeddings mapping natural language queries to FHIR APIs across patient records. **HIPAA-compliant RBAC**: role-based access controls for PHI - built compliance-first. **Human-in-the-loop**: clinician-facing interface for record retrieval with transparent query translation. **~60% retrieval time reduction.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

### CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

## EDUCATION

**M.S. Data Analytics Engineering** - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

**B.Tech Computer Science & Engineering** - Sree Venkateshwara College of Engineering 2013-2017

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

**AI & Agents:** LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, LLM-as-Judge Evaluation, CI/CD Eval Gates, Vector Search

**NLP & ML:** BERT, spaCy, Hugging Face, PyTorch, TensorFlow, scikit-learn, Embeddings, Feature Engineering, NumPy, Pandas, Statistical Modeling

**Healthcare & Compliance:** FHIR APIs, HIPAA-compliant RBAC, PHI Encryption (Libsodium), Secure Coding, Clinical NLP

**Cloud & DevOps:** AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), FastAPI, Flask, React, PostgreSQL, MySQL, RabbitMQ

**Process:** TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum